

TOPAS-nBio v4.1.0 (OpenTOPAS v4.2.1, Geant4-11.3.2)

Regression testing (cf. TOPAS-nBio v4.0 (OpenTOPAS v4.0, Geant4-11.1.3))

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Introduction

Welcome to the TOPAS-nBio regression test results!

This document depicts the results for 13 separate regression tests, comparing TOPAS-nBio v4.1.0 to TOPAS-nBio v4.0.

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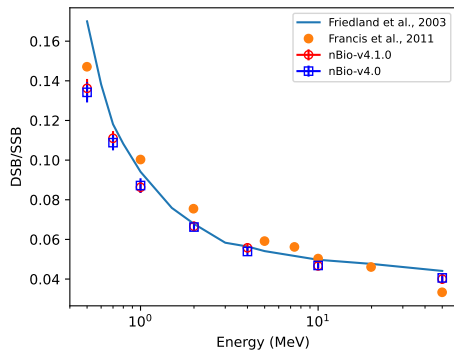
Nanodosimetry III: TsEmDNAPhysics

Nanodosimetry III: g4em-dna_opt2

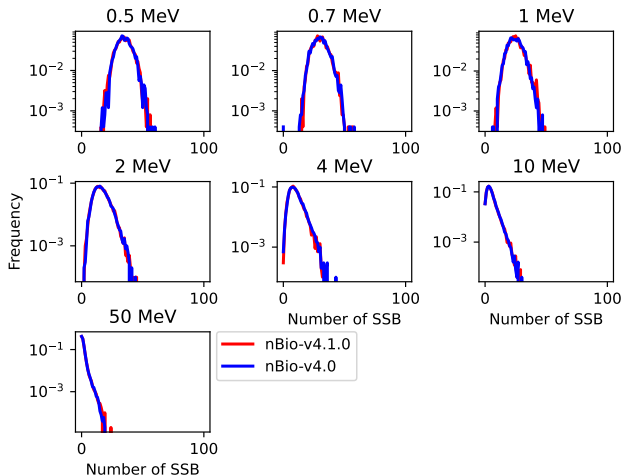
Nanodosimetry III: g4em-dna_opt4

Nanodosimetry III: g4em-dna_opt6

DBSCAN - TsEmDNAPhysics

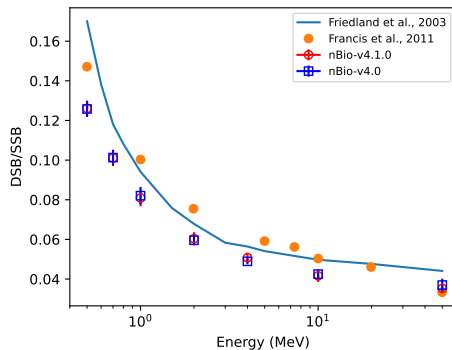


	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	371.2 +/- 5.9	391.1 +/- 6.8

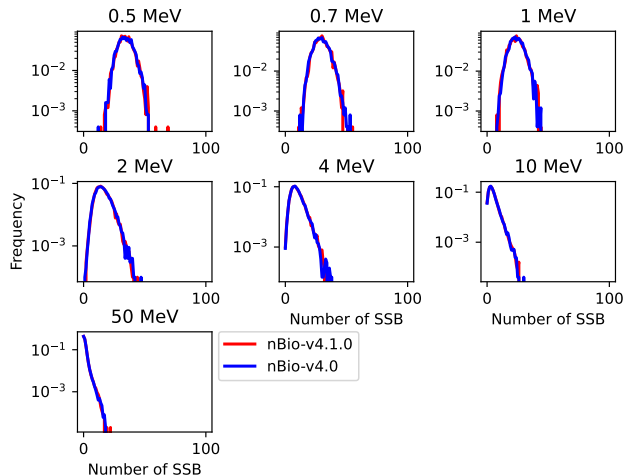


- Francis Z, Villagrasa C, Clairand I. Simulation of DNA damage clustering after proton irradiation using an adapted DBSCAN algorithm. *Comput Methods Programs Biomed.* 2011; 101(3):265-270. doi:10.1016/j.cmpb.2010.12.012

DBSCAN - g4em-dna_opt2

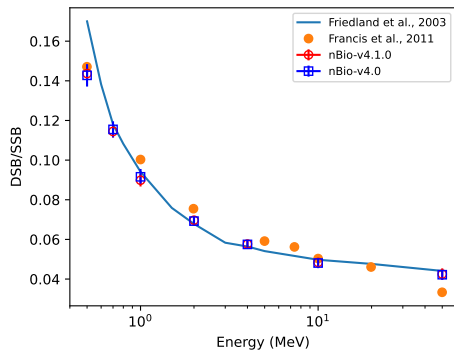


	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	373.0 +/- 6.8	387.0 +/- 2.5

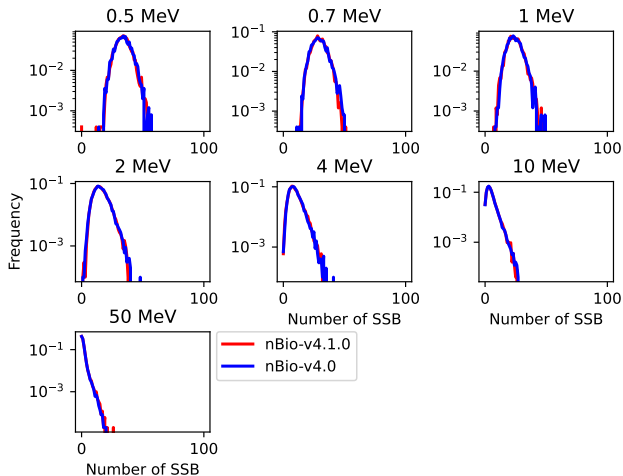


- Francis Z, Villagrasa C, Clairand I. Simulation of DNA damage clustering after proton irradiation using an adapted DBSCAN algorithm. *Comput Methods Programs Biomed.* 2011; 101(3):265-270. doi:10.1016/j.cmpb.2010.12.012

DBSCAN - g4em-dna_opt4

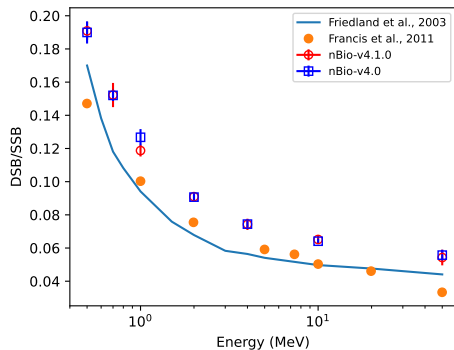


	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	822.1 +/- 6.0	845.7 +/- 11.4

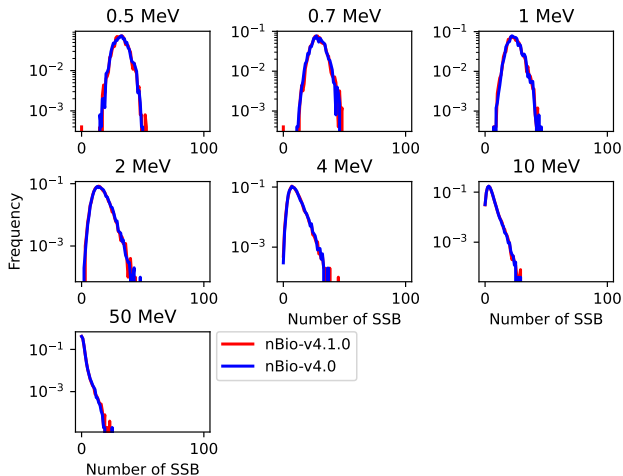


- Francis Z, Villagrasa C, Clairand I. Simulation of DNA damage clustering after proton irradiation using an adapted DBSCAN algorithm. *Comput Methods Programs Biomed.* 2011; 101(3):265-270. doi:10.1016/j.cmpb.2010.12.012

DBSCAN - g4em-dna_opt6

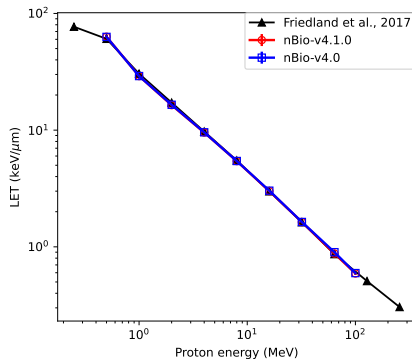


	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	327.0 +/- 10.0	298.1 +/- 4.9

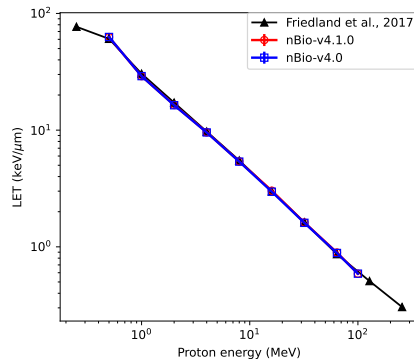


- Francis Z, Villagrasa C, Clairand I. Simulation of DNA damage clustering after proton irradiation using an adapted DBSCAN algorithm. *Comput Methods Programs Biomed.* 2011; 101(3):265-270. doi:10.1016/j.cmpb.2010.12.012

LET I



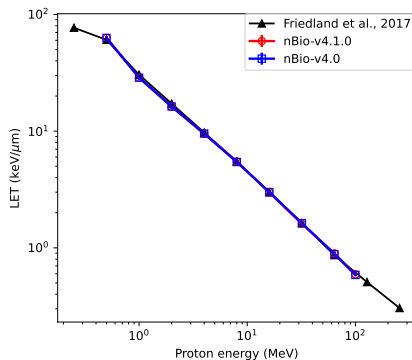
	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	481.4 +/- 10.9	493.1 +/- 4.4



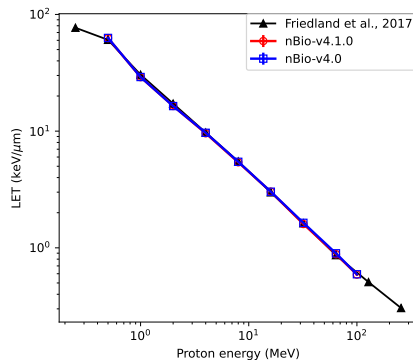
	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	473.0 +/- 5.2	501.1 +/- 6.0

► LET as a function of proton energy for TsEmDNAPhysics (left) and g4em-dna_opt2 (right).

LET II



	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	912.2 +/- 17.3	976.8 +/- 7.6

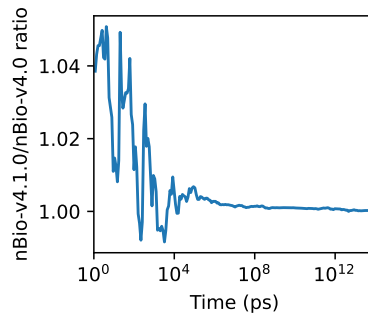
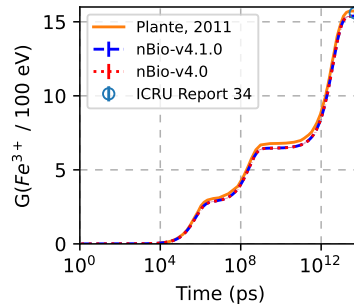


	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	492.2 +/- 9.1	493.6 +/- 4.7

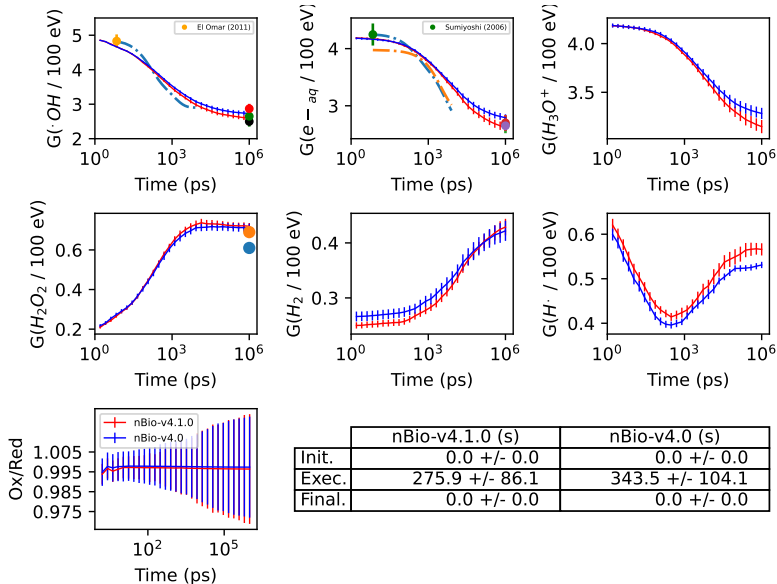
▶ LET as a function of proton energy for g4em-dna_opt4 (left) and g4em-dna_opt6 (right).

Fricke: IRT

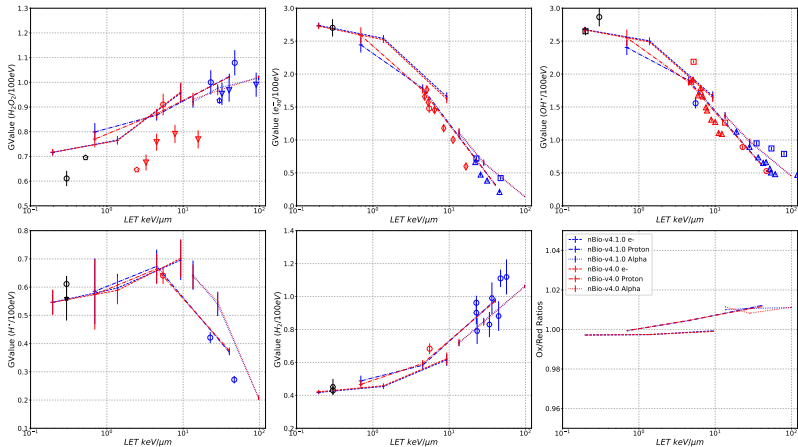
	nBio-v4.1.0	nBio-v4.0
Exec. (s)	112.696 +/- 2.836	111.144 +/- 2.924
Value (/100eV)	15.386 +/- 0.006	15.382 +/- 0.010



G-value: step-by-step

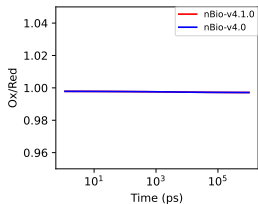
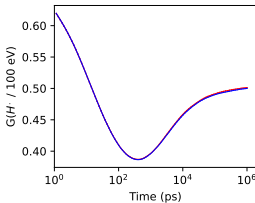
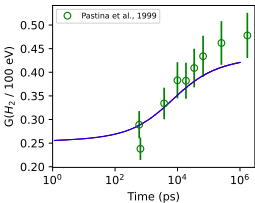
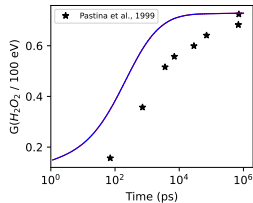
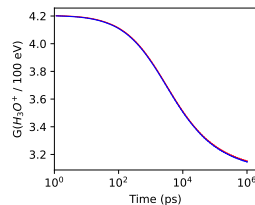
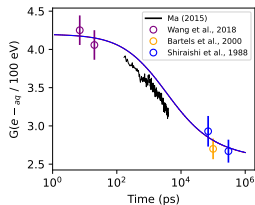
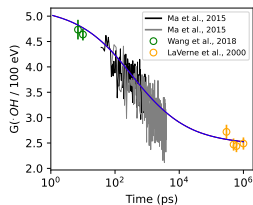


G-value vs. LET: step-by-step



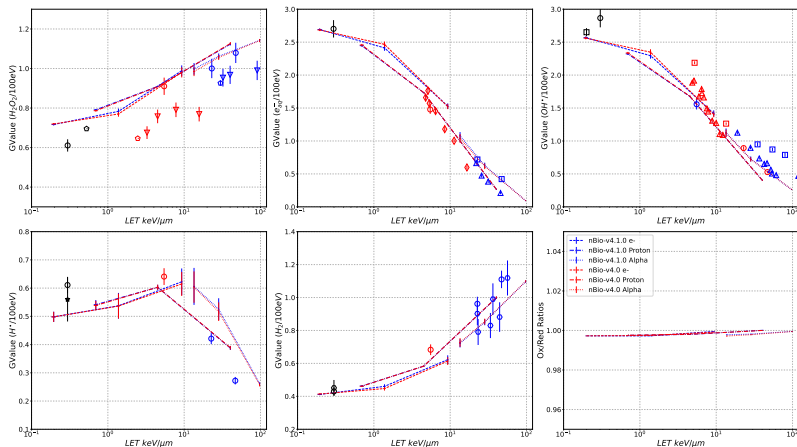
	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	2739.107 +/- 107.952	2900.672 +/- 87.720

G-value: IRT



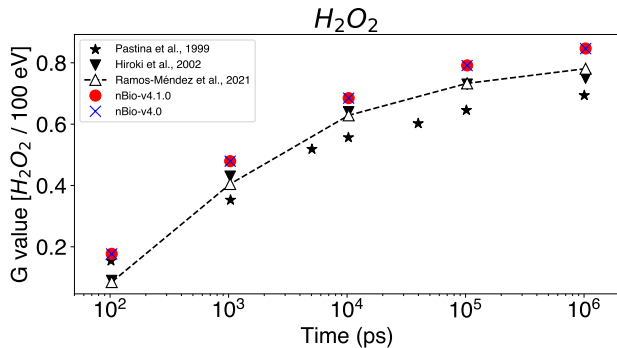
	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	123.620 +/- 3.813	121.276 +/- 4.566

G-value vs. LET: IRT



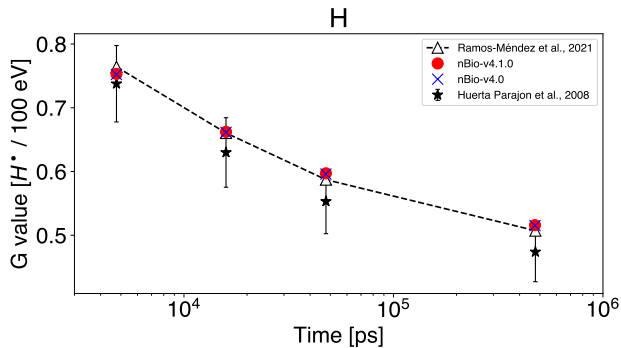
	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	123.614 +/- 2.965	127.835 +/- 7.107

G-value of H_2O_2 : IRT



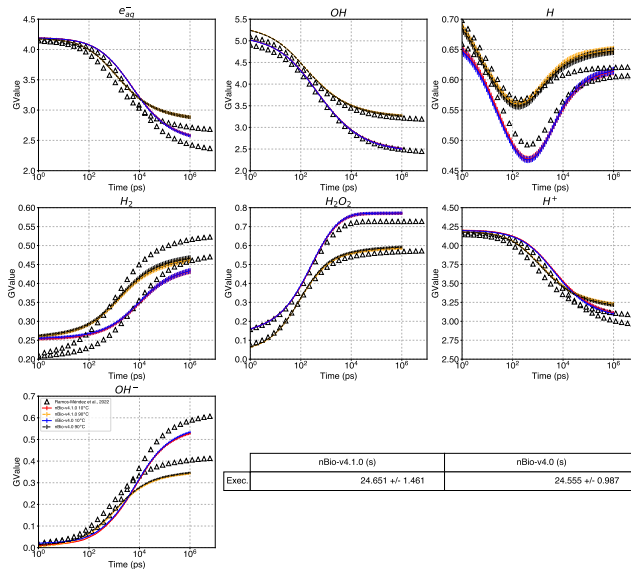
	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	91.440 +/- 2.474	89.405 +/- 1.838

G-value of H: IRT



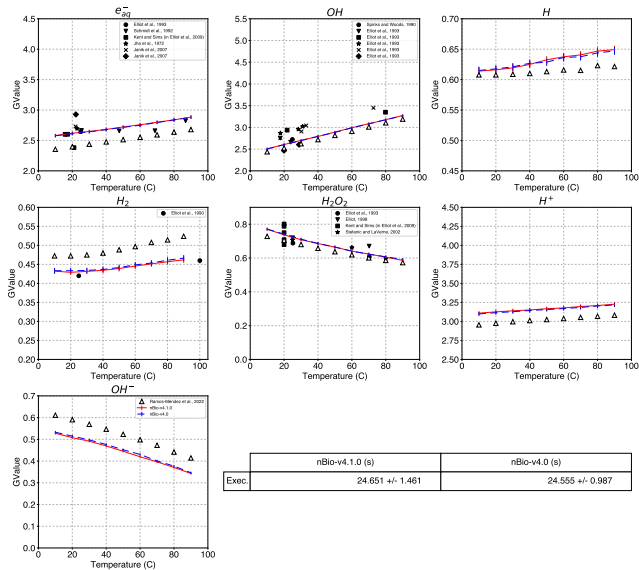
	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	256.001 +/- 9.333	256.620 +/- 8.557

G-value and Temperature I: IRT



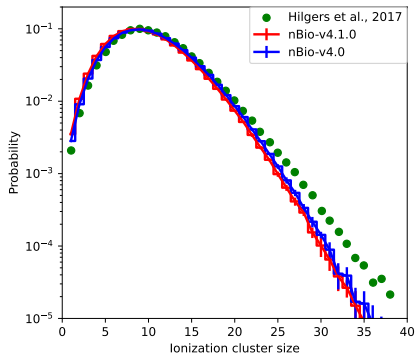
	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	24.651 +/- 1.461	24.555 +/- 0.987

G-value and Temperature II: IRT

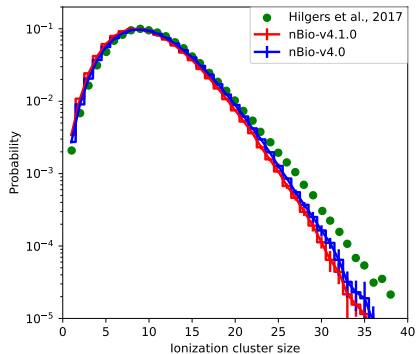


	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	24.651 +/- 1.461	24.555 +/- 0.987

Nanodosimetry I: TsEmDNAPhysics and g4em-dna_opt2

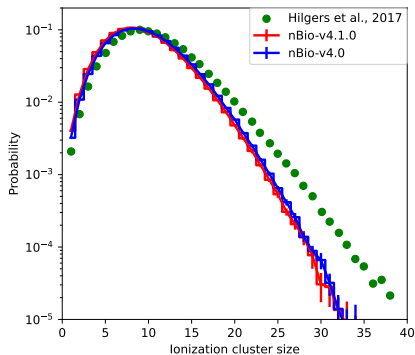


	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	15838.2 +/- 112.8	16270.4 +/- 209.9

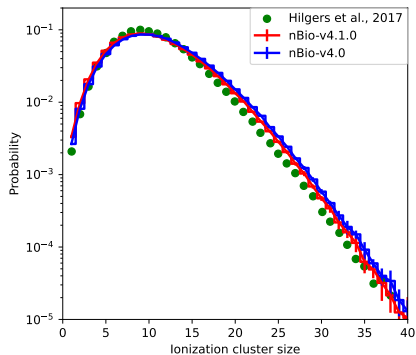


	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	15998.9 +/- 466.9	16879.5 +/- 335.3

Nanodosimetry I: g4em-dna_opt4 and g4em-dna_opt6

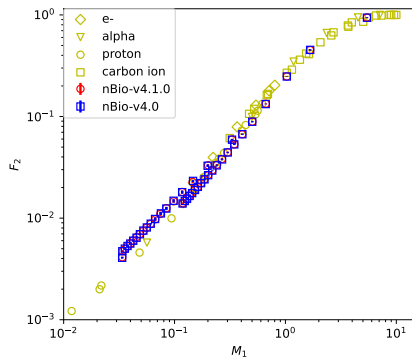


	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	11944.3 +/- 190.7	12295.8 +/- 133.5

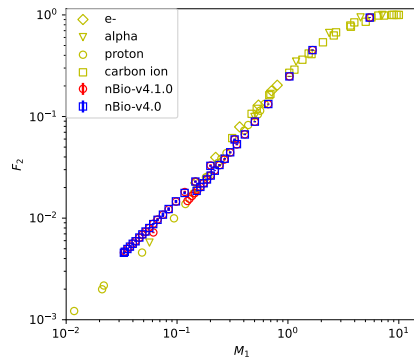


	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	13109.0 +/- 398.9	12059.8 +/- 190.0

Nanodosimetry II: TsEmDNAPhysics and g4em-dna_opt2



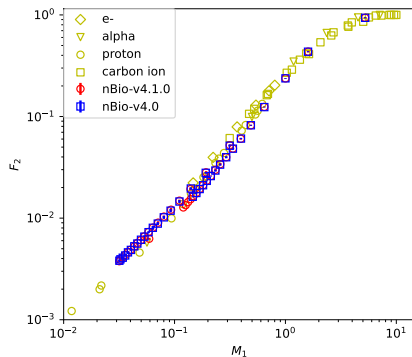
	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	2540.5 +/- 11.0	1777.3 +/- 20.7



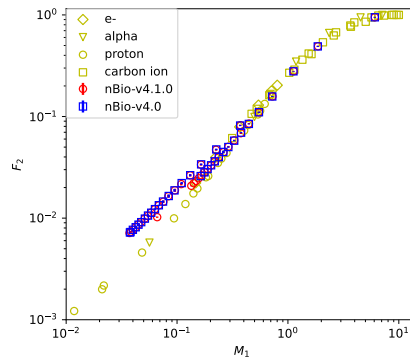
	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	2484.2 +/- 28.4	1488.7 +/- 12.8

Conte V, Selva A, Colautti P, et al., Nanodosimetry: Towards a new concept of radiation quality. *Radiat Prot Dosimetry*. 2018;180(1-4):150-156. doi:10.1093/rpd/ncx175

Nanodosimetry II: g4em-dna_opt4 and g4em-dna_opt6



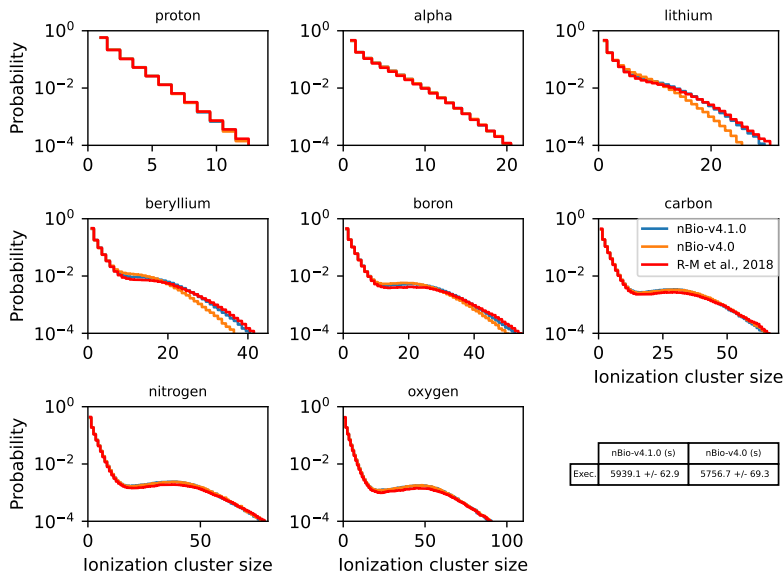
	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	39831.4 +/- 734.6	27024.8 +/- 454.1



	nBio-v4.1.0 (s)	nBio-v4.0 (s)
Exec.	33819.0 +/- 379.6	20844.0 +/- 25.6

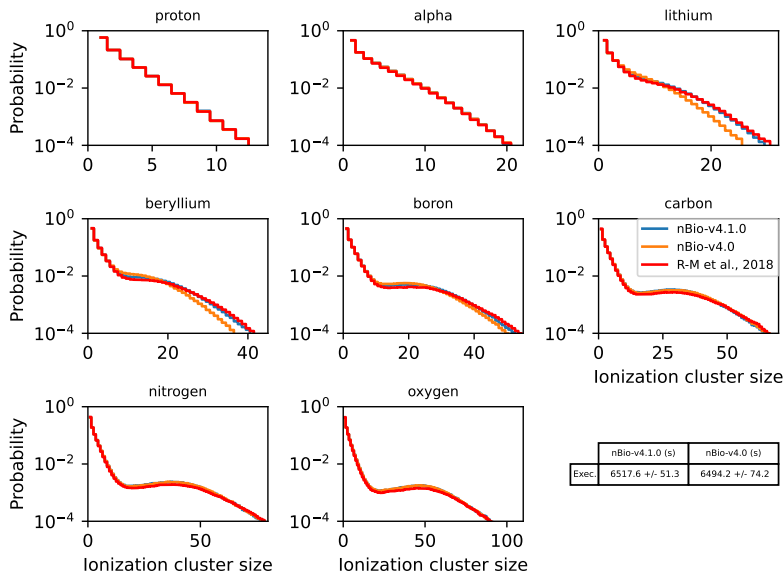
Conte V, Selva A, Colautti P, et al., Nanodosimetry: Towards a new concept of radiation quality. *Radiat Prot Dosimetry*. 2018;180(1-4):150-156. doi:10.1093/rpd/ncx175

Nanodosimetry III: TsEmDNAPhysics



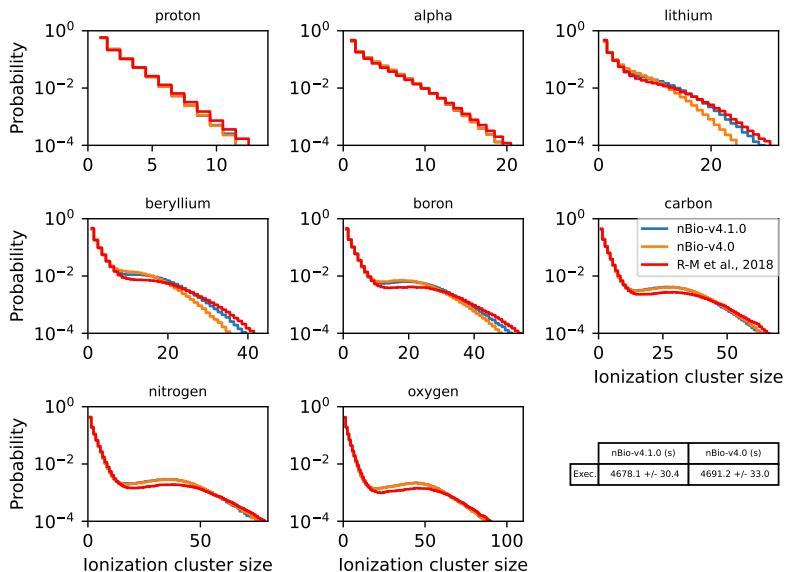
Ramos-Méndez J, Burigo LN, Schulte R, Chuang C, Faddegon B. Fast calculation of nanodosimetric quantities in treatment planning of proton and ion therapy. *Phys Med Biol.* 2018;63(23):235015. doi:10.1088/1361-6560/aaeeee

Nanodosimetry III: g4em-dna_opt2



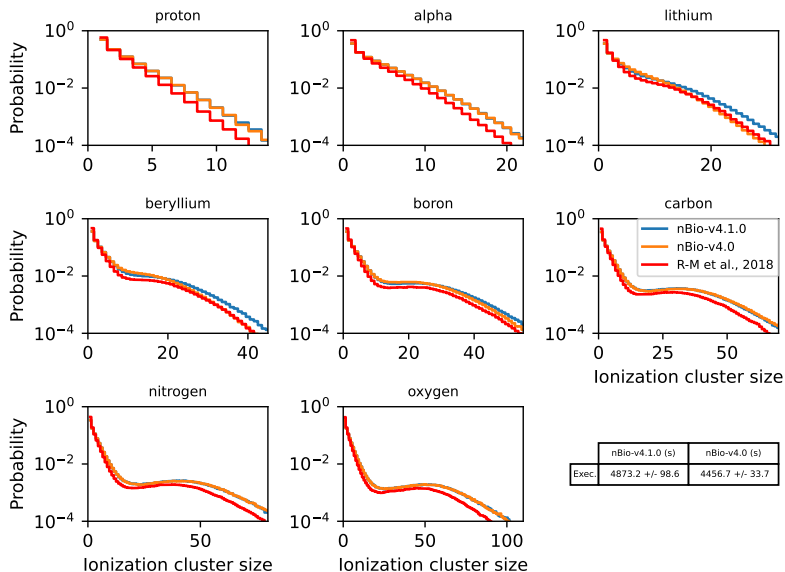
Ramos-Méndez J, Burigo LN, Schulte R, Chuang C, Faddegon B. Fast calculation of nanodosimetric quantities in treatment planning of proton and ion therapy. *Phys Med Biol.* 2018;63(23):235015. doi:10.1088/1361-6560/aaeeee

Nanodosimetry III: g4em-dna_opt4



Ramos-Méndez J, Burigo LN, Schulte R, Chuang C, Faddegon B. Fast calculation of nanodosimetric quantities in treatment planning of proton and ion therapy. *Phys Med Biol.* 2018;63(23):235015. doi:10.1088/1361-6560/aaeeee

Nanodosimetry III: g4em-dna_opt6



Ramos-Méndez J, Burigo LN, Schulte R, Chuang C, Faddegon B. Fast calculation of nanodosimetric quantities in treatment planning of proton and ion therapy. *Phys Med Biol.* 2018;63(23):235015. doi:10.1088/1361-6560/aaeeee